8086 Assembly Programs (DOS INT 21h)

Part A: Basic DOS I/O Programs

Program 1: Input a character and display it on the next line

Task: Display a prompt, read one character from the keyboard, print it on the next line, then exit to DOS.

```
; Program 1
.MODEL SMALL
.STACK 100H
.DATA

.CODE
MAIN PROC
    ; Display the prompt '?'
    MOV AH, 02H            ; DOS: display character
    MOV DL, '?'
    INT 21H

    ; Read one character (echoed). Result in AL
    MOV AH, 01H            ; DOS: input character
    INT 21H
    MOV BL, AL             ; Save input

    ; New line (CR + LF)
    MOV AH, 02H
    MOV DL, 0DH
    INT 21H
    MOV DL, 0AH
    INT 21H

    ; Display the entered character
    MOV AH, 02H
    MOV DL, BL
    INT 21H

    ; Exit to DOS
    MOV AX, 4C00H
    INT 21H
MAIN ENDP
END MAIN
```

Program 2: Display the message HELLO!

Task: Initialize DS, print a $-terminated string using DOS function 09H, then exit.

```
; Program 2
.MODEL SMALL
.STACK 100H

.DATA
MSG DB 'HELLO!$'

.CODE
MAIN PROC
    ; Initialize DS
    MOV AX, @DATA
    MOV DS, AX

    ; Print the message
    LEA DX, MSG
    MOV AH, 09H            ; DOS: print string ($-terminated)
    INT 21H

    ; Exit to DOS
    MOV AX, 4C00H
    INT 21H
MAIN ENDP
END MAIN
```

Program 3: Convert a lowercase letter to uppercase

Assumption: Input is a lowercase letter (a-z). Conversion uses ASCII rule: uppercase = lowercase - 20H.

```
; Program 3
.MODEL SMALL
.STACK 100H

.DATA
MSG1 DB 'ENTER A LOWERCASE LETTER: $'
MSG2 DB 0DH,0AH,'IN UPPERCASE IT IS: $'

.CODE
MAIN PROC
    ; Initialize DS
    MOV AX, @DATA
    MOV DS, AX
```

```
    ; Prompt
    LEA DX, MSG1
    MOV AH, 09H
    INT 21H

    ; Input character (AL)
    MOV AH, 01H
    INT 21H

    ; Convert to uppercase (ASCII)
    SUB AL, 20H

    ; Print message
    LEA DX, MSG2
    MOV AH, 09H
    INT 21H

    ; Print converted letter
    MOV DL, AL
    MOV AH, 02H
    INT 21H

    ; Exit
    MOV AX, 4C00H
    INT 21H
MAIN ENDP
END MAIN
```

Program 4: Input two integers (single digits), add, and display the result (single digit)

Note: Inputs are single digits (0-9). The result, which is also (0-9), is printed.

```
Prompts the user to enter two integers.

Adds the two integers together.

Displays the result of the addition.

.MODEL SMALL            ; Define the memory model as small (code and data share
the same segment)
.STACK 100H             ; Allocate 256 bytes (100H) for the stack

.DATA                   ; Data segment begins here
MSG1 DB 'ENTER FIRST NUMBER: $'  ; Prompt message for the first number
MSG2 DB 0DH,0AH,'ENTER SECOND NUMBER: $' ; Prompt message for the second number
MSG3 DB 0DH,0AH,'THE RESULT IS: $'    ; Message to display the result
NUM1 DB ?               ; Reserve a byte for the first number input
NUM2 DB ?               ; Reserve a byte for the second number input
RESULT DB ?             ; Reserve a byte to store the result of the addition

.CODE                   ; Code segment begins here
MAIN PROC
    ; Initialize DS (data segment register)
    MOV AX,@DATA         ; Load the address of the data segment into AX
```

```
    MOV DS,AX            ; Move the address from AX into DS to initialize DS

    ; Print prompt to enter the first number
    LEA DX,MSG1          ; Load the effective address of MSG1 into DX (first
number prompt)
    MOV AH,9             ; DOS function 9: Display string ending with '$'
    INT 21H              ; Call DOS interrupt to display the first number prompt

    ; Input the first number
    MOV AH,1             ; DOS function 1: Input a character (first number)
    INT 21H              ; Call DOS interrupt to get the first number (in AL)
    SUB AL,30H           ; Convert ASCII to integer by subtracting '0' (ASCII
value of '0' is 30H)
    MOV NUM1,AL          ; Store the first number in NUM1

    ; Print prompt to enter the second number
    LEA DX,MSG2          ; Load the effective address of MSG2 into DX (second
number prompt)
    MOV AH,9             ; DOS function 9: Display string ending with '$'
    INT 21H              ; Call DOS interrupt to display the second number prompt

    ; Input the second number
    MOV AH,1             ; DOS function 1: Input a character (second number)
    INT 21H              ; Call DOS interrupt to get the second number (in AL)
    SUB AL,30H           ; Convert ASCII to integer by subtracting '0'
    MOV NUM2,AL          ; Store the second number in NUM2

    ; Add the numbers
    MOV AL,NUM1          ; Load the first number into AL
    ADD AL,NUM2          ; Add the second number (NUM2) to AL
    MOV RESULT,AL        ; Store the result in RESULT

    ; Display the result
    LEA DX,MSG3          ; Load the effective address of MSG3 into DX (result
message)
    MOV AH,9             ; DOS function 9: Display string ending with '$'
    INT 21H              ; Call DOS interrupt to display the result message

    ; Display the result (convert the result from integer to ASCII)
    ADD RESULT,30H       ; Convert the result from integer to ASCII (add '0' which
is 30H)
    MOV DL,RESULT        ; Load the result into DL (character to display)
    MOV AH,2             ; DOS function 2: Display a character
    INT 21H              ; Call DOS interrupt to display the result

    ; DOS exit
    MOV AH,4CH           ; DOS function 4Ch: Exit the program and return control
to DOS
    INT 21H              ; Call DOS interrupt to terminate the program

MAIN ENDP                ; End of main procedure
END MAIN                 ; End of the program
```

Part B: Expression Evaluation Programs (single-digit inputs and output)

Note: A and B read as single digits (0-9). The result is also single-digit (0-9).

Problem 1: Calculate 3A + 2B

Expression: 3A + 2B

```
.MODEL SMALL
.STACK 100H
.DATA
MSG1 DB 'ENTER A: $'
MSG2 DB 0DH,0AH,'ENTER B: $'
MSG3 DB 0DH,0AH,'3A + 2B = $'
A DB ?
B DB ?
RESULT DB ?
.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX

    LEA DX,MSG1
    MOV AH,9
    INT 21H

    MOV AH,1
    INT 21H
    SUB AL,30H
    MOV A,AL

    LEA DX,MSG2
    MOV AH,9
    INT 21H

    MOV AH,1
    INT 21H
    SUB AL,30H
    MOV B,AL

    MOV AL,A
    ADD AL,A
    ADD AL,A
    MOV BL,B
    ADD BL,B
    ADD AL,BL
    MOV RESULT,AL

    LEA DX,MSG3
    MOV AH,9
    INT 21H

    ADD RESULT,30H
    MOV DL,RESULT
    MOV AH,2
```

```
    INT 21H

    MOV AH,4CH
    INT 21H
MAIN ENDP
END MAIN
```

Problem 2: Calculate A * B - C

Expression: A * B - C

```
.MODEL SMALL
.STACK 100H
.DATA
MSG1 DB 'ENTER A: $'
MSG2 DB 0DH,0AH,'ENTER B: $'
MSG3 DB 0DH,0AH,'ENTER C: $'
MSG4 DB 0DH,0AH,'A × B - C = $'
A DB ?
B DB ?
C DB ?
RESULT DB ?
.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX

    LEA DX,MSG1
    MOV AH,9
    INT 21H

    MOV AH,1
    INT 21H
    SUB AL,30H
    MOV A,AL

    LEA DX,MSG2
    MOV AH,9
    INT 21H

    MOV AH,1
    INT 21H
    SUB AL,30H
    MOV B,AL

    LEA DX,MSG3
    MOV AH,9
    INT 21H

    MOV AH,1
    INT 21H
    SUB AL,30H
    MOV C,AL
```

```
    MOV AL,A
    MUL B
    SUB AL,C
    MOV RESULT,AL

    LEA DX,MSG4
    MOV AH,9
    INT 21H

    ADD RESULT,30H
    MOV DL,RESULT
    MOV AH,2
    INT 21H

    MOV AH,4CH
    INT 21H
MAIN ENDP
END MAIN
```

Problem 3: Calculate 2A + B^2

Expression: 2A + B^2

```
.MODEL SMALL
.STACK 100H
.DATA
MSG1 DB 'ENTER A: $'
MSG2 DB 0DH,0AH,'ENTER B: $'
MSG3 DB 0DH,0AH,'2A + B² = $'
A DB ?
B DB ?
RESULT DB ?
.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX

    LEA DX,MSG1
    MOV AH,9
    INT 21H

    MOV AH,1
    INT 21H
    SUB AL,30H
    MOV A,AL

    LEA DX,MSG2
    MOV AH,9
    INT 21H

    MOV AH,1
    INT 21H
    SUB AL,30H
    MOV B,AL
```

```
    MOV AL,A
    ADD AL,A
    MOV BL,B
    MUL BL
    ADD AL,BL
    MOV RESULT,AL

    LEA DX,MSG3
    MOV AH,9
    INT 21H

    ADD RESULT,30H
    MOV DL,RESULT
    MOV AH,2
    INT 21H

    MOV AH,4CH
    INT 21H
MAIN ENDP
END MAIN
```

Problem 4: Calculate A + 2B + 3C

Expression: A + 2B + 3C

```
.MODEL SMALL
.STACK 100H
.DATA
MSG1 DB 'ENTER A: $'
MSG2 DB 0DH,0AH,'ENTER B: $'
MSG3 DB 0DH,0AH,'ENTER C: $'
MSG4 DB 0DH,0AH,'A+2B+3C = $'
A DB ?
B DB ?
C DB ?
RESULT DB ?
.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX

    LEA DX,MSG1
    MOV AH,9
    INT 21H
    MOV AH,1
    INT 21H
    SUB AL,30H
    MOV A,AL

    LEA DX,MSG2
    MOV AH,9
    INT 21H
    MOV AH,1
```

```
    INT 21H
    SUB AL,30H
    MOV B,AL

    LEA DX,MSG3
    MOV AH,9
    INT 21H
    MOV AH,1
    INT 21H
    SUB AL,30H
    MOV C,AL

    MOV AL,A
    MOV BL,B
    ADD BL,B
    ADD AL,BL
    MOV BL,C
    ADD BL,C
    ADD BL,C
    ADD AL,BL
    MOV RESULT,AL

    LEA DX,MSG4
    MOV AH,9
    INT 21H

    ADD RESULT,30H
    MOV DL,RESULT
    MOV AH,2
    INT 21H

    MOV AH,4CH
    INT 21H
MAIN ENDP
END MAIN
```

Problem 5: Calculate A^2 - B^2

Expression: A^2 - B^2

```
.MODEL SMALL
.STACK 100H
.DATA
MSG1 DB 'ENTER A: $'
MSG2 DB 0DH,0AH,'ENTER B: $'
MSG3 DB 0DH,0AH,'A² - B² = $'
A DB ?
B DB ?
RESULT DB ?
.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX
```

```
    LEA DX,MSG1
    MOV AH,9
    INT 21H
    MOV AH,1
    INT 21H
    SUB AL,30H
    MOV A,AL

    LEA DX,MSG2
    MOV AH,9
    INT 21H
    MOV AH,1
    INT 21H
    SUB AL,30H
    MOV B,AL

    MOV AL,A
    MUL A
    MOV BL,AL
    MOV AL,B
    MUL B
    MOV CL,AL
    MOV AL,BL
    SUB AL,CL
    MOV RESULT,AL

    LEA DX,MSG3
    MOV AH,9
    INT 21H

    ADD RESULT,30H
    MOV DL,RESULT
    MOV AH,2
    INT 21H

    MOV AH,4CH
    INT 21H
MAIN ENDP
END MAIN
```